AY: 2025-26

Class:	**TE**	**Semester:**	**VI**
Course Code:	**CSL603**	**Course Name:**	**SEPM**

Name of Student:	**Shravani S. Raut**
Roll No.:	**40**
Experiment No.:	**07**
Title of the Experiment:	**To Setup and Run Selenium Tests in Jenkins Using Maven**
Date of Performance:	**25/02/2026**
Date of Submission:	**11/03/2026**

Evaluation

Performance Indicator	Max. Marks	Marks Obtained
Performance	5	
Understanding	5	
Journal work and timely submission	10	
Total	20	

Performance Indicator	Exceed Expectations (EE)	Meet Expectations (ME)	Meet Expect Below Expectations (BE)
Performance	4-5	2-3	1
Understanding	4-5	2-3	1
Journal work and timely submission	8-10	5-8	1-4

Checked by

Name of Faculty: Ms Rujuta Vartak
Signature:
Date:

Experiment No-07

Aim: To Setup and Run Selenium Tests in Jenkins Using Maven

Objective: Objective is to setup enables seamless integration of automated testing into the CI/CD pipeline, facilitating faster feedback loops and promoting a culture of continuous improvement in software development.

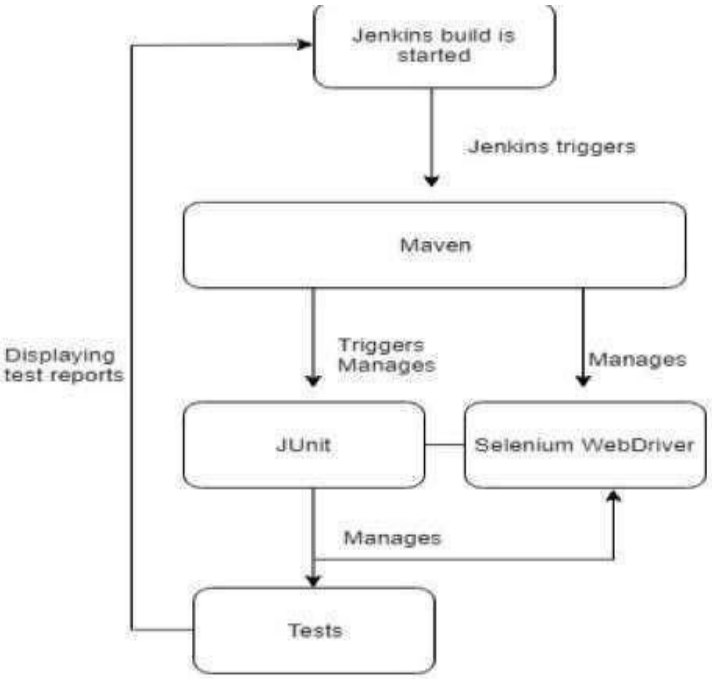
Theory:

Jenkins is the leading open-source continuous integration tool developed by Hudson lab. It is cross- platform and can be used on Windows, Linux, Mac OS and Solaris environments. Jenkins is written in Java. It has taken the place as one of the best open-source tools that allow continuous integration and build management.

Running Selenium tests in Jenkins allows you to run your tests every time your software changes and deploy the software to a new environment when the tests pass. Jenkins can schedule your tests to run at specific time. You can save the execution history and Test Reports. Jenkins supports Maven for building and Testing a project in continuous integration

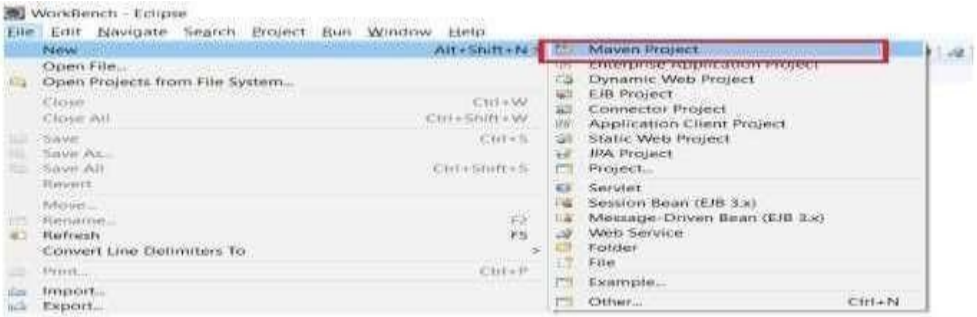
Maven is a powerful project / build management tool, based on the concept of a POM (Project Object Model) that includes project information and configuration information for Maven such as construction directory, source directory, dependency, test source directory, Goals, plugins, etc.

Integrating Maven with Selenium provides following benefits Apache Maven provides support for managing the full lifecycle of a test project. Maven is used to define project structure, dependencies, build, and test management. Using pom.xml(Maven) you can configure dependencies needed for building testing and running code. Maven automatically downloads the necessary files from the repository while building the project.

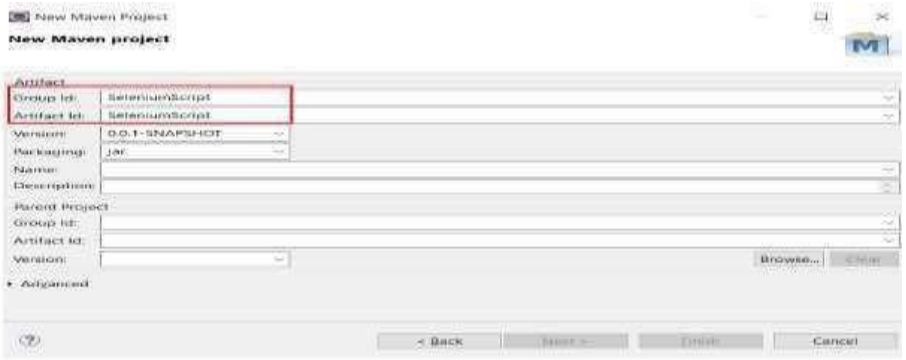


Steps: -Create a Maven Selenium script---

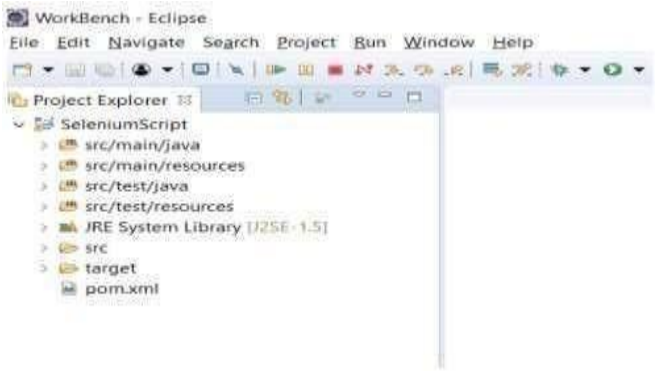
1. In Eclipse IDE, create a new project by selecting **File | New | Maven Project** from Eclipse menu.



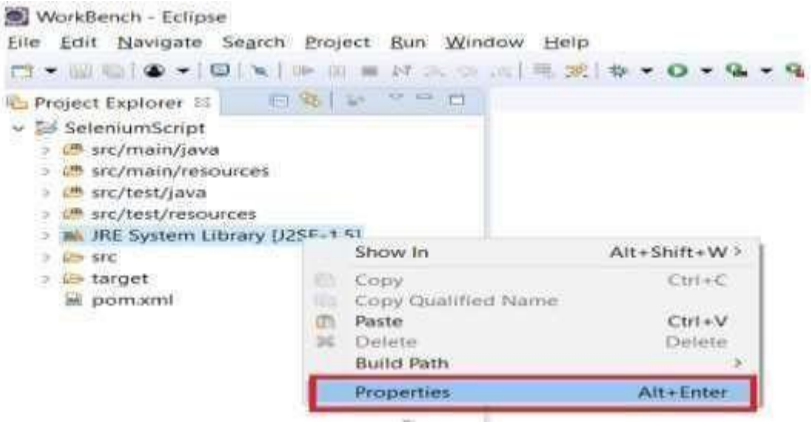
2. On the **New Maven Project** dialog select the **Create a simple project** and click **Next**



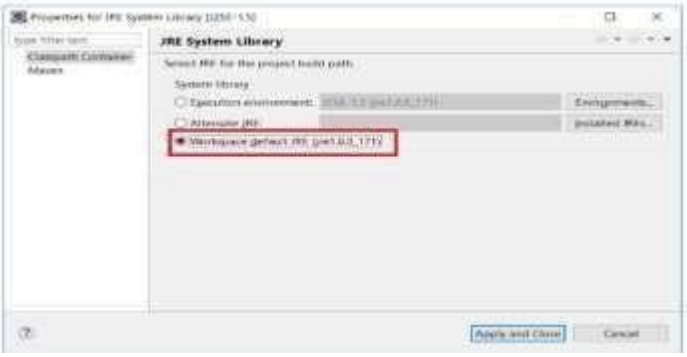
3. Enter **SeleniumScript** in **Group Id**: and **Artifact Id**: and click finish



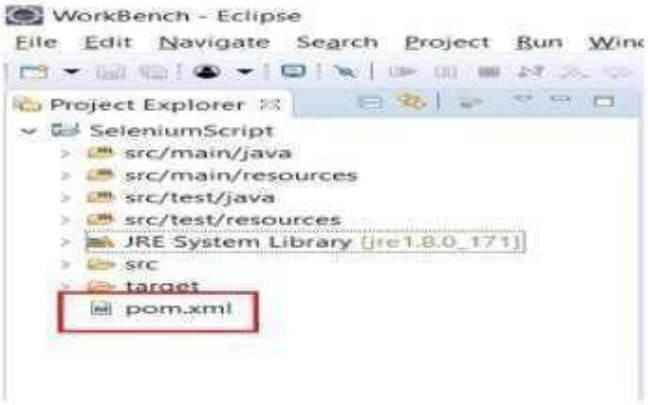
4. Eclipse will create webdriverTest.
5. Right click on JRE System Library and select the Properties option from the menu.



6. On the Properties for JRE System Library dialog box , make sure Workspace default JRE is selected and click ok.



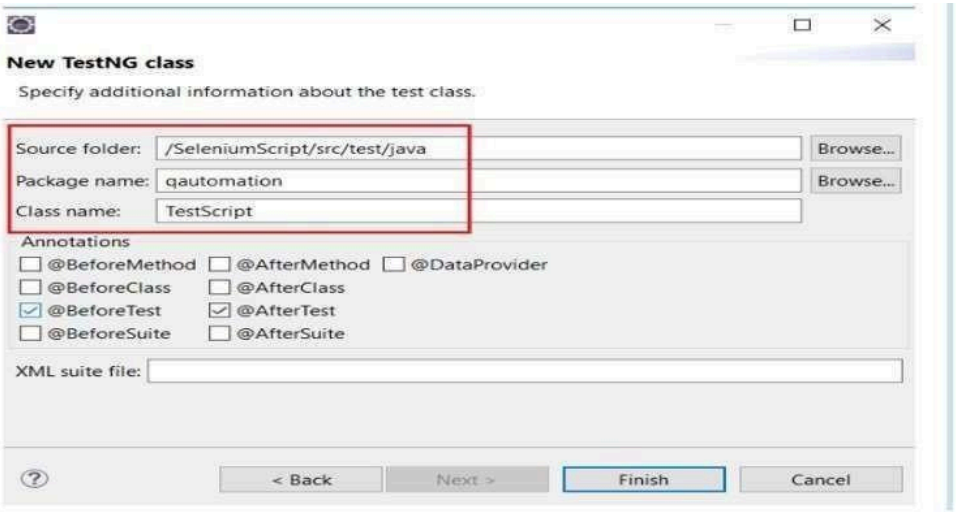
7. Select pom.xml from project explorer.



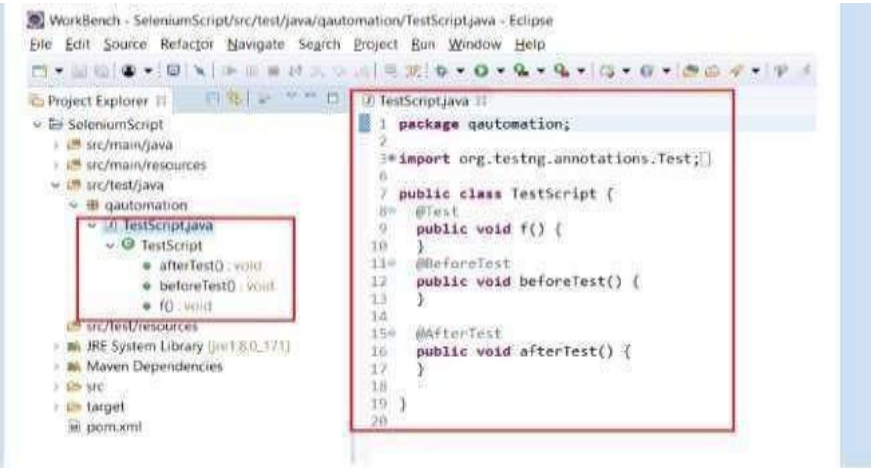
8. Add selenium, Maven, TestNG, Junit dependencies to pom.xml in the code.

```
<dependencies>
        <dependency>
            <groupId>junit</groupId>
            <artifactId>junit</artifactId>
            <version>3.8.1</version>
            <scope>test</scope>
        </dependency>
        <dependency>
            <groupId>org.seleniumhq.selenium</groupId>
            <artifactId>selenium-java</artifactId>
            <version>2.45.0</version>
                </dependency>
        <dependency>
            <groupId>org.testng</groupId>
            <artifactId>testng</artifactId>
            <version>6.8</version>
            <scope>test</scope>
        </dependency>
</dependencies>
```

9. Create a new file TestNG class File|New|Others|TestNG|TestNG Class. Enter Package name as "Qautomation" and "TestScript" in the Name:textbox and click on the Finish button.



10. Eclipse will create the TestScript class



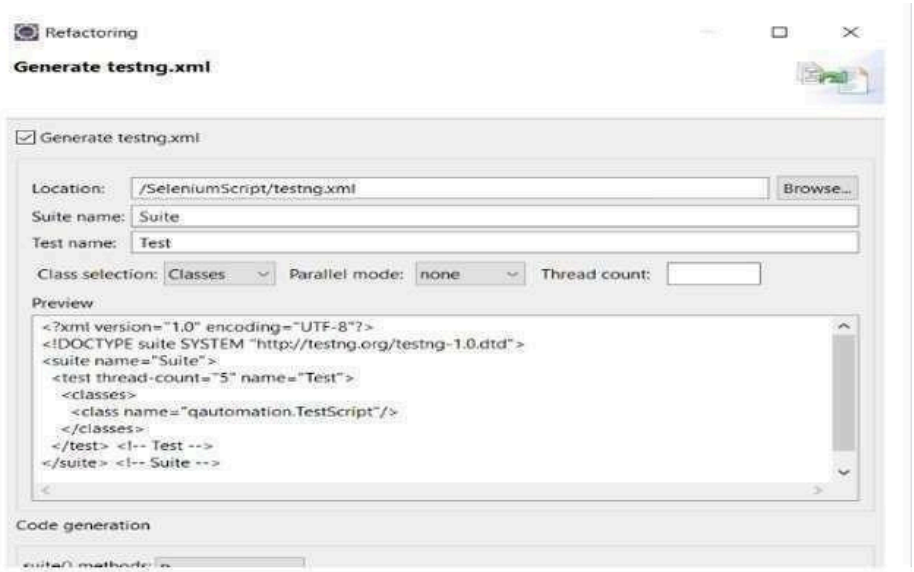
11. Add following code to the TestScript class and respective browser drivers for chrome,firefox and IE.

```
Import package qautomation; import org.testng.annotations.Test;
import org.testng.annotations.BeforeT est; import java.util.HashMap;
import java.util.Map;
import java.util.concurrent.TimeUnit;
import org.openqa.selenium.WebDriver;
import org.openqa.selenium.chrome.ChromeDriver;
import org.openqa.selenium.chrome.ChromeOption;
org.openqa.selenium.remote.DesiredCapabilit ies;
import org.testng.Assert;
import org.testng.annotations.AfterTes t; public class TestScript {
        public static WebDriver driver=null; public String browser =
        System.getProperty("browser"); public String url =
        System.getProperty("URL");
 @BeforeTest
  public void beforeTest() {
        if(browser.equalsIgnoreCase("Chrome"))
        {
        System.setProperty("webdriver.chrome.driver",
        System.getProperty("user.dir")+"\\chromedriver.e xe");
        Map<String, Object> prefs = new HashMap<String, Object>();
        ChromeOptions options = new ChromeOptions();
        options.setExperimentalOption("prefs", prefs);
        options.addArguments("--disable-arguments");
        options.addArguments("--test-type");
        options.addArguments("test");
        options.addArguments("disable-infobars");
        driver = new ChromeDriver(options);
        }
        else if(browser.equalsIgnoreCase("FireFox"))
        {
        System.setProperty(FirefoxDriver.SystemProperty.DRIVER_USE_M
        ARIONET TE,"true");
        System.setProperty(FirefoxDriver.SystemProperty.BROWSER_LOG
        FILE,  S yste m.getProperty("user.dir")+"\\FireFoxLogs.txt");
        System.setProperty("webdriver.gecko.driver",
        System.getProperty("user.dir")+"\\geckodriver_v23.exe");
        FirefoxProfile profile = new FirefoxProfile();
        profile.setAcceptUntrustedCertificates(false); FirefoxOptions
        options = new FirefoxOptions().setProfile(profile); driver = new
        FirefoxDriver(options);
```

```
            driver.manage().timeouts().implicitlyWait(20,
            TimeUnit.SECONDS); driver.manage().window().maximize();
            }
      else if (browser.equalsIgnoreCase("IE"))
            {
            System.setProperty("webdriver.ie.driver",
            System.getProperty("user.dir")+"\\IEDriverServer351.e
            xe"); DesiredCapabilities caps =
            DesiredCapabilities.internetExplorer()
      caps.setCapability(InternetExplorerDriver.INT RODUCE_FLAKINESS_BY_IGN O
      RING_SECURITY_DOMAINS,true);
      caps.setCapability(InternetExplorerDriver.IGN ORE_ZOOM_SETTING,true);
      caps.setCapability(InternetExplorerDriver.UNE XPECTED_ALERT_BEHAVIOR
      ," accept"); caps.setCapability(InternetExplorerDriver.RE
      QUIRE_WINDOW_FOCUS,true);
      caps.setCapability(InternetExplorerDriver.INIT IAL_BROWSER_URL,"http://ww w.
      google.com/");
      driver = new InternetExplorerDriver(caps);
      driver.manage().timeouts().implicitl yWait(20, TimeUnit.SECONDS);
      driver.manage().window().maximiz e();
      }
      driver.manage().timeouts().implicitl yWait(20, TimeUnit.SECONDS);
      driver.manage().window().maximiz e();

      public void
      TestApplication() {
      driver.get(url);
      String title = driver.getTitle();
      System.out.println("Title="+title);
      Assert.assertTrue(title.contains("QAutomation
      "));
       }
      @AfterTest
       public void afterTest() {
              driver.quit();
       }
}
```

12. Right click on the WebdriverTest and select TestNG| Convert to TestNG. Eclipse will create testing.xml which says that you need to run only one test with the name

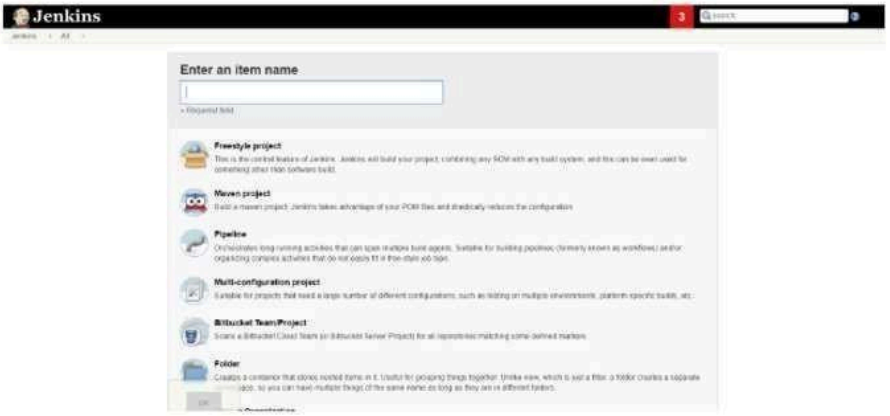


13. Adding dependencies and plugins Additionally we need to add
    1. Maven-compiler-plugin
    2. Maven-surefire-plugin
    3. Testng.xml

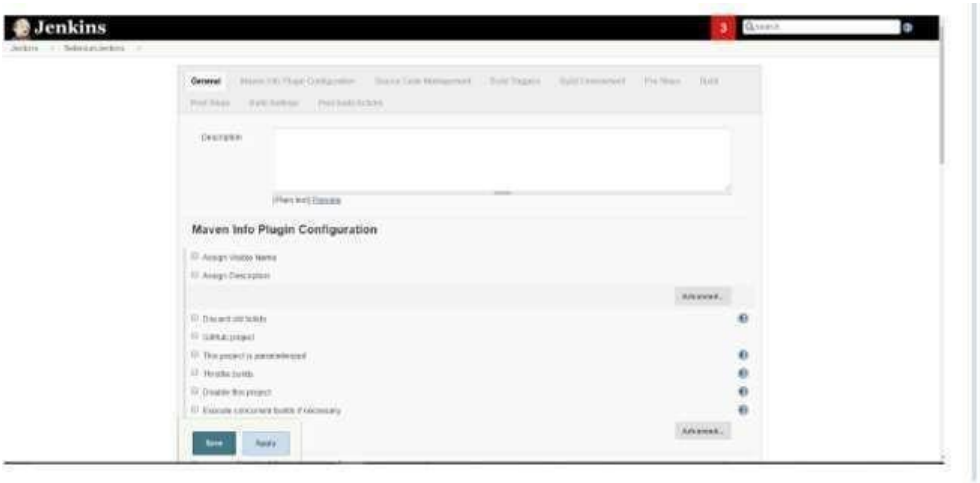
---------Integrating your test to Jenkins----------

1. Launch and login into jenkins URL – http://localhost:8080/

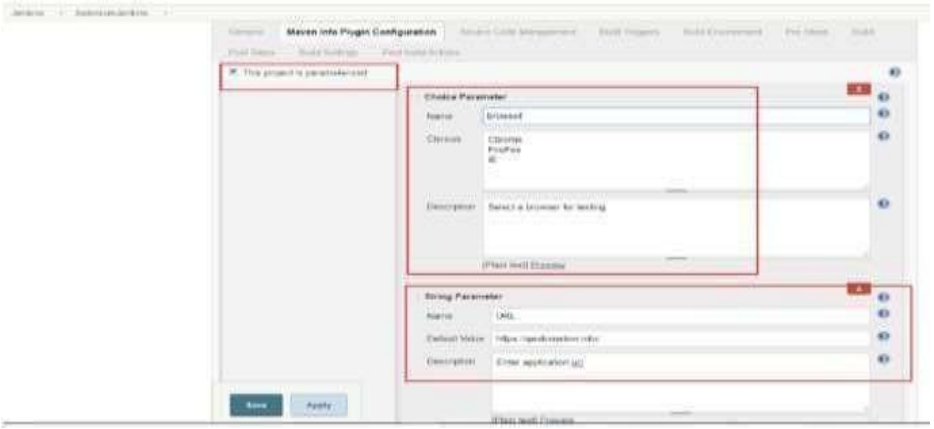
2. Click on new item and enter an appropriate name for the new job , select Maven Project and click on save.



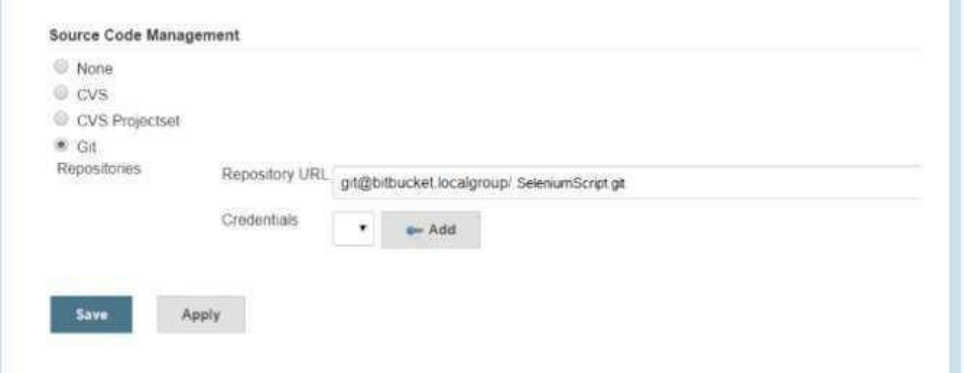
3. A new empty job has been created at this point.



4. Jenkins Parameterized Build in Jenkins just check the checkbox **This project is paramerized** and add the parameter by **Add Parameter** as per your project requirement.



5. If code is located on Git Under **Source Management** , select the appropriate repository for the location of project and pass the URL and credentials.



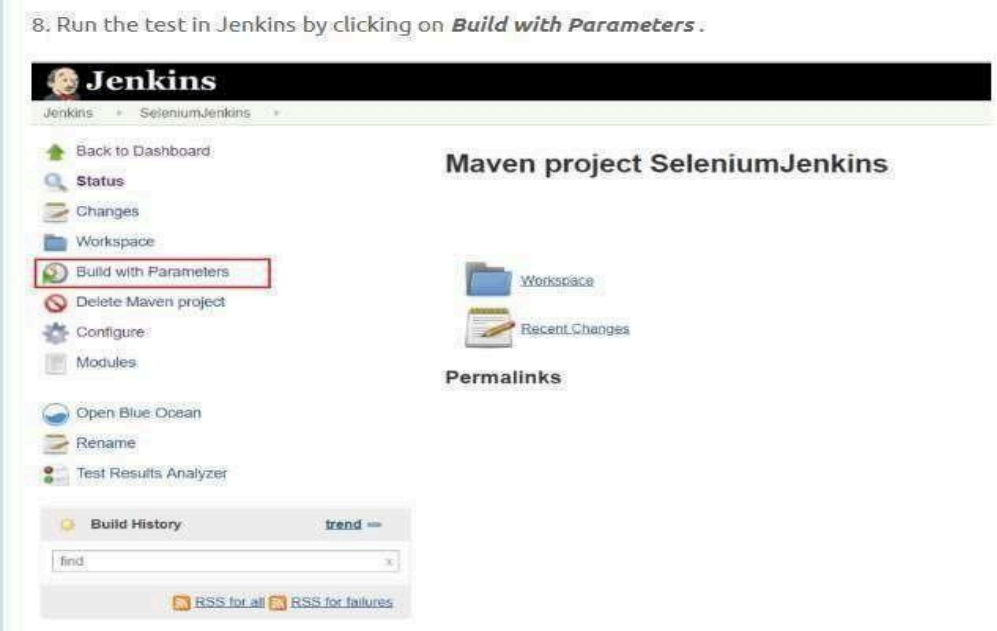
6. In the "pre-steps" build section another set of parameters can be passed to the Jenkins build. Specify the Maven targets that need to be executed in order to run test.

if your source code is located on Git the do below setting under **Build** section:

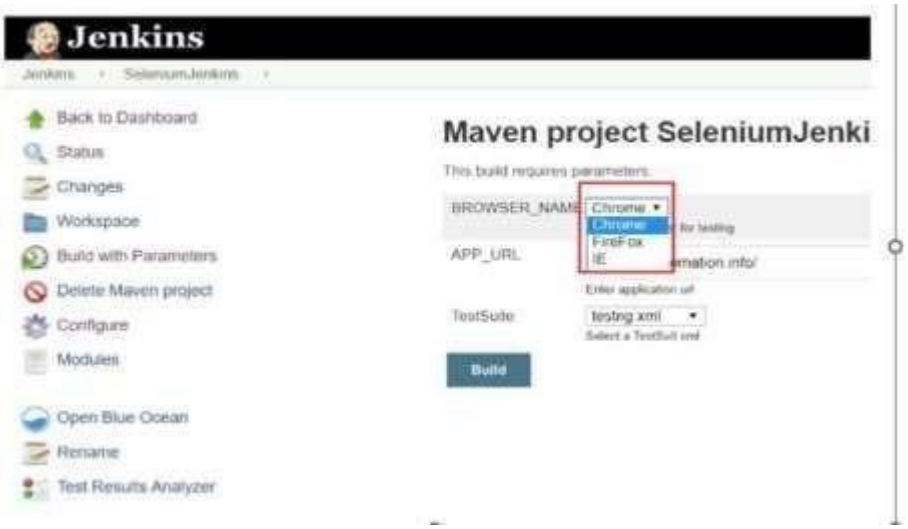
If you have selenium code on your local just pass the pom.xml path in ***Root POM***.

7. Run the test in Jenkins by clicking on Building with Parameters.

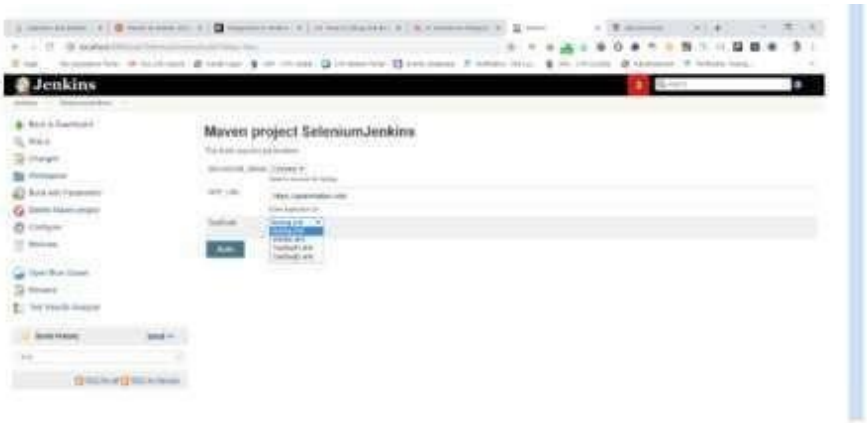
8. Run the test in Jenkins by clicking on ***Build with Parameters***.



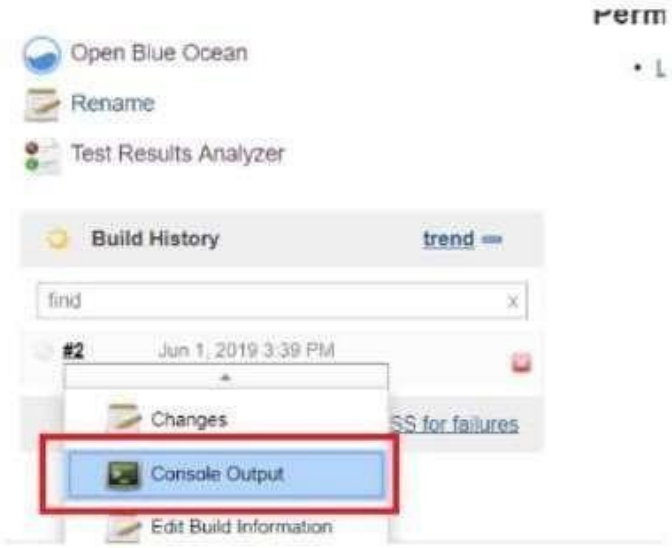
8. Select the browser you want to run from dropdown.



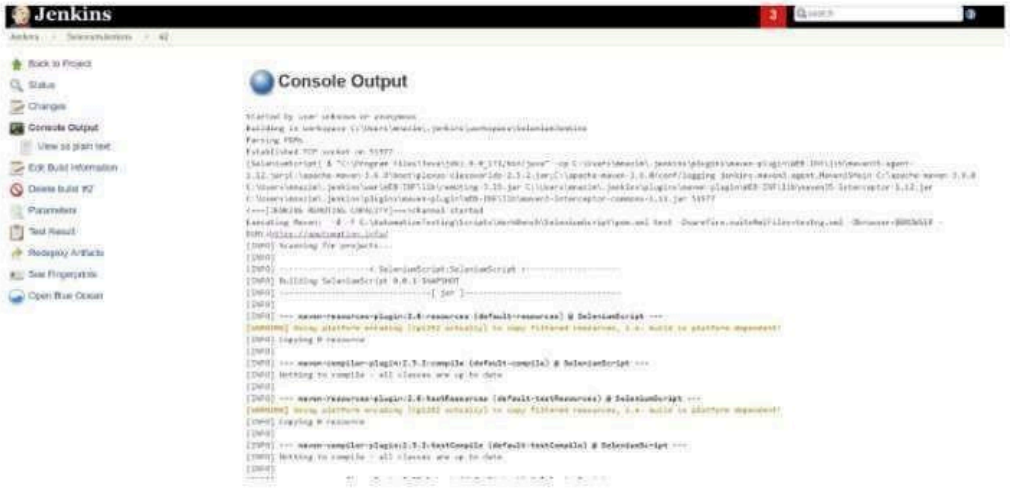
9. Select the TestSuit file.



10. Click the build button and go to console output .

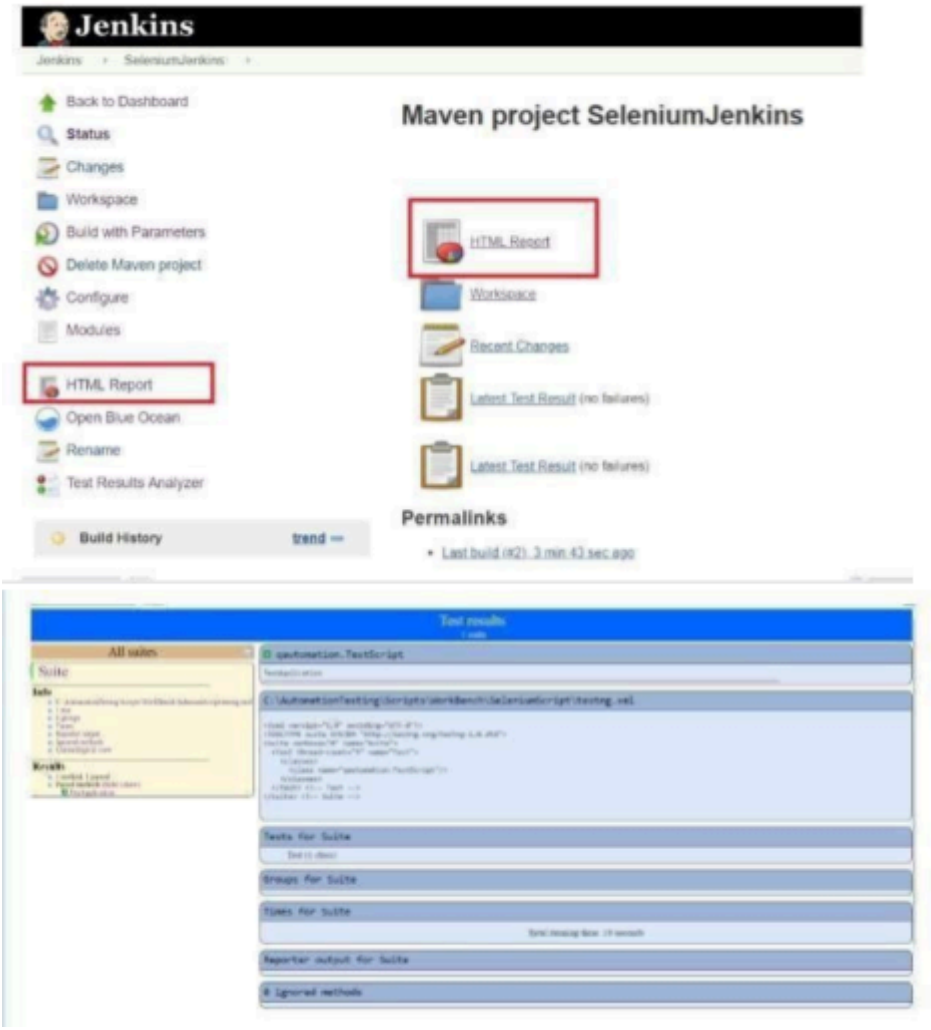


11. See the logs from **Console Output** window.

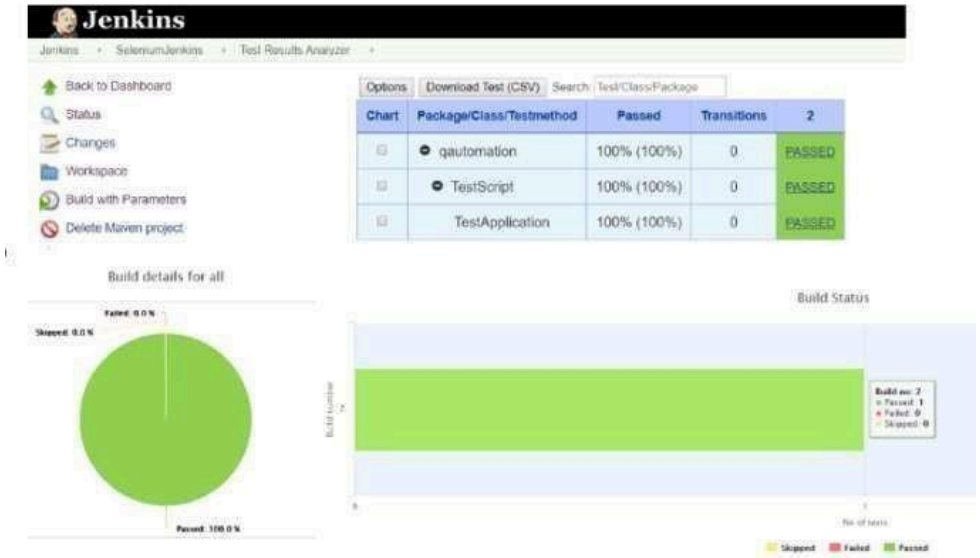


Note : Blue color of ball of console output is that build is successful

12. View the html report just click on the link.



13. Click Test Analyzer to analyse the result.



Conclusion:

Q1. Which browsers are supported by selenium webdriver?

Selenium WebDriver supports all major modern web browsers through specific driver implementations. The officially supported browsers include:

- Google Chrome: Automated using ChromeDriver.
- Mozilla Firefox: Automated using GeckoDriver.
- Microsoft Edge: Automated using Microsoft Edge Driver (specifically the Chromium-based version).
- Apple Safari: Automated using SafariDriver, which is built into macOS.
- Opera: Automated using OperaDriver (based on Chromium).
- Internet Explorer (IE): Automated using IEDriverServer, though it is largely legacy and replaced by Edge in modern workflows.
- Headless Browsers: Supports invisible browsers like HtmlUnit for fast, GUI-less execution.

Selenium also provides frameworks like Appium for mobile browser testing on Android and iOS devices.

Q2. What are some features of selenium 4?

Selenium 4 introduces several key enhancements designed to improve stability, simplify script creation, and provide deeper browser control.

Key Features of Selenium 4

1. **W3C WebDriver Standardization**: Selenium 4 fully adopts the W3C WebDriver protocol, ensuring consistent test execution across all major browsers (Chrome, Firefox, Safari, and Edge) without needing the legacy JSON Wire Protocol.
2. **Relative Locators:** Also known as "Friendly Locators," this feature allows you to find web elements based on their visual position relative to others using commands like above(), below(), toLeftOf(), toRightOf(), and near().
3. **Chrome DevTools Protocol (CDP) Support:** Native CDP integration allows testers to perform advanced tasks such as network interception, capturing performance metrics, and mocking geolocations directly from their scripts.
4. **Redesigned Selenium Grid:** The Selenium Grid is now more scalable and supports Docker for easier containerized testing. It also features a revamped, user-friendly UI for monitoring sessions.
5. **Improved Window and Tab Management:** A new newWindow() API simplifies creating and switching between multiple windows or tabs within a single test session without manual handle management.

6. **Upgraded Selenium IDE:** The Selenium IDE has been rebuilt as a browser extension for Chrome and Firefox, featuring a better control flow (if/else, while loops) and the ability to export recorded tests to multiple programming languages.
7. **Enhanced Documentation:** The official Selenium documentation has been completely revamped with a clearer structure and updated code examples for all supported languages.
8. **Deprecation of Desired Capabilities:** The old DesiredCapabilities class is deprecated in favor of specific browser options classes like ChromeOptions and FirefoxOptions for more organized browser configuration